MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS ALUMINUM (CP), TYPE I, CLASS 2, COMPOSITION D

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Fe+Si	Al	Cu	Zn	Others (1)	
				Each	Total
1.0 max.	99.0 min.	0.2 max.	0.1 max.	0.05 max.	0.15 max.

(1) Determination not required for routine acceptance.

Particle Size/Distribution

+140 Mesh (+106 µm)	1.0 max.
+170 Mesh (+90 µm)	7.0 max.
+200 Mesh (+75 µm)	50.0 max.
+325 Mesh (+45 µm)	89.0 min.

Sprayed Coating Properties

Bond Strength	8.3 MPa (1200 psi) min.
Hardness	Rockwell H, 25 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 15% max. Unmelted Particles - 15% max.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Deposit Efficiency and Deposition Rate

80% min for N_2/H_2 @ 6.5 lb/hr
70% min for A/H_2 @ 6.5 lb/hr

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

General Electric Company	B50TF57, Class A
Pratt & Whitney Aircraft	PWA 1320
Rolls-Royce, Ltd.	MSRR 9507/13

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/05)

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